



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/641558

Date:

01/12/2022(17:17:02)

Subject: Review of LMI "GUIDE LINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER" with Ref No: LMI/OIN/OPS/MECH/032

REF.: SMPP/O&M/MM/PP/

Date: 01.12.2022

Subject: Revision of LMI "GUIDE LINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER" with Ref No: LMI/OIN/OPS/MECH/032

-

The above approved LMI was revised in month of Aug '2022. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

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On: 01/12/2022(17:17:02) **Ref:** SMTTP/MECH MAINT-/2022-23/LMI/641558

The subject LMI has been revised as per changes envisaged in COS-ISO-00-OGN/OPS/MECH/053.

The changes are incorporated in LMI doc with ref no: LMI/OGN/OPS/MECH/053
ISSUE NO.1; REVISION NO. 02; DATE: DEC'2022.

CA may please accord approval for this revised version of LMI.

Submitted by:

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Changes incorporated in revised LMI in following clauses:

1. Clause 7 (A)- 2,3,4,5
2. Clause 7 (B)- 2,6,12,15,16
3. Clause 7 (C) - 2,3,5,9,10
4. Clause 7 (E)- 3

The above changes have been highlighted in attached document "New LMI doc"

Submitted for approval

Submitted by:

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Submitted for approval pls

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May please be accorded approval.

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On:20/12/2022(22:47:11) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI/641558

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/053

ISSUE NO.1; REVISION NO. 02; DATE: DEC'2022

**GUIDELINES FOR CARRYING OUT SAFE
OPERATION OF SKY CLIMBERS INSIDE
BOILER**

APPROVED FOR

***Implementation*.....**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	GUIDELINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/053/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV: 02; DEC'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/MECH/053/ Rev 0</u>
<u>4.PREPARED BY :</u> SR. MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY :</u> DGM (MM-PP)	SIGNATURE-----
AGM (BMD)	SIGNATURE-----
AGM (SAFETY)	SIGNATURE-----
AGM (FQA)	SIGNATURE-----
GM (MTC)	SIGNATURE-----
GM (O & M)	SIGNATURE-----
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: GUIDELINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER REV 02; DATE: DEC'2022		
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<u>2.0 INTRODUCTION</u> Sky Climbers are suspended access platforms essential for inspection and repair of Pressure Parts of Boiler predominantly in 1st Pass. As it involves operation of suspended platform up to 75m (approx.) height inside boiler, fall from height is major risk involved. Hence, safety is a key aspect while using the Sky Climbers. It is necessary that each and every person who intends to utilize the Sky Climber familiarize himself about the proper and safe methods relating to use of Sky Climbers along with their functions and limitations. The objective of this LMI is to improve the safety of sky climber operation by following specific guidelines. <u>3.0 SCOPE</u> The scope of this LMI is to detail the checks to be adopted while using sky climbers, maintenance practices to be followed, training to be imparted to the personnel involved and records to be maintained. <u>4.0 REFERENCE DOCUMENT</u> Operation Guidance Note: COS-ISO-00-OGN/OPS/MECH/053 Rev.0, circulated by operation services - NTPC Corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> LMI/OIN/OPS/MECH/032/ ISSUE-1; REV: 0.00; JAN'2020 <u>6.0 SPECIFIC INSTRUCTIONS</u> The objective of the LMI is sought to be achieved through following actions: i) By carrying out specific checks and instructions at different stages of using sky climber i.e. before sky climber assembly, before & during operation and in times of emergency ii) By implementation of time bound maintenance and Overhaul strategy		

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<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u> <u>A) GUIDELINES FOR OPERATORS SAFETY DURING SKY CLIMBER OPERATION</u> 1. Soft copy of OEM / OES manual should be kept in station boiler dashboard and should be referred by person involved in erecting, dismantling and using Sky Climber. One hard copy of OEM/OES manual, all safety instructions and list of trained personnel should be kept on Sky Climber. 2. All the trained persons who service, install, dismantle, or use the suspended access equipment must fully comprehend and act in accordance with OEM / OES manual. 3. A trained person is a designated person, suitably trained by OEM, qualified by knowledge and practical experience to carry out installation, operation and dismantling of Sky Climber. 4. Competent persons must be emotionally strong and physically able to withstand the stress of working at elevations. Health checkup is complete along with height test. No previous history of Vertigo/Epilepsy/Height phobia. 5. All persons using Sky Climber must at all-time wear PPE's like safety belts, safety shoes, Helmet etc. Safety belt should be attached to the side member of Sky climber. Induction safety training and job specific training like work at height and PPE are complete. 6. All persons must be imparted toolbox talk before start of work and briefed about the risk associated with work. 7. Maintain communication with the supervisor at ground level continuously. Responsibility - HOD (BMD) <u>B) CHECK POINTS BEFORE SKY CLIMBER ASSEMBLY</u> 1. Inspect and test the Sky Climber before use, rigging, re-rigging and after de- rigging and at regular intervals to make sure that it is maintained and in safe workable condition. 2. The supervisor should assign a competent person for inspection purpose. 3. Check wire rope physically every time before using it in Sky Climber. It should not be deformed such as kinked, birdcage, worn, flattened, slackening, thinning, loop formation, loose wires or damaged. In case of any damage is observed, discard the wire rope.		

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4. The wire rope if found corroded, with its surface completely roughened or pitted, or if its wires are slack within the strands due to wastage, then wire rope should be replaced. The wire rope at roof anchor point arrangement should have 3 No. U clamps along with rubber thimbles/rubber pads. 5. The wire rope may be replaced if wire rope diameter has decreased by more than the prescribed limit (10% as recommended by OEM). 6. Check physically condition of Motor & Gear Box along with gear box oil level. 7. Check Safety lock and test it by inserting one side of rope into it. It should operate satisfactorily. 8. Check physically condition of platforms and other members. It should not be damaged. 9. Check edges of guide pipe and clamping of wire rope. 10. Check dead weight is attached to safety rope and it is hanging. 11. Check power cable and short cable. It should be ok. 12. An additional bulb of 24 V and a torch should be available on the platform. 13. Check for adequate light inside the boiler. 14. Check bottom hopper is sealed properly. 15. One additional set of sky climber should be mobilized for emergency handling. 16. Clinker should be completely removed in the direction of sky climber movement. Responsibility - HOD (BMD) <u>C) CHECK POINTS BEFORE SKY CLIMBER OPERATION</u> 1. All suspended access equipment must be handled with care during handling of equipment, installation and use and to avoid any damage of its components. 2. When welding is carried out from a suspended platform, precautions should be taken to ensure that there is no possibility of making an earth leakage path through the suspension steel wire ropes for the electric current because of faulty welding leads. This might damage the steel wire ropes. This can be avoided by insulating the platform and the steel wire ropes from the structure or by using insulating thimbles/rubber/insulating sheets over steel wire ropes above sky climber equipment. 3. Operate and check all the Safety locks thoroughly by operating the reset lever manual. 4. When 3 phase power supply is used, all checks related to power supply of the hoist are to be carried out with the essential use of RCCB of 30 mA capacity. Healthiness of RCCB to be ensured before use.		

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5. Check operation of Motor, G Box and Brake. There should not be any abnormal sound or slippage.(Take platform few meters above ground & come back) 6. Check all bolted connections on platform, roof anchorage visually per shift or per day before use or as per OEM manual. 7. Check all switches and controls as per OEM manual. 8. Check the working of emergency lowering system of hoists. 9. Check both ends of sky climber move in a synchronous manner. 10. Check during gas cutting/welding , molten slag does not fell over sky climber wire rope/rubber wall wheels/power cables/lighting cable. 11. Check platform is not overloaded. (Not more than 5 persons) 12. Check operators are competent, trained and experienced. 13. Check for an additional Safety lock in center. 14. Controls are operating properly. 15. Check all persons are using PPE like safety helmet, safety belt and safety shoe. 16. A protocol should be signed by Agency, BMD, FQA and Safety department before operation of Sky Climber. Responsibility - HOD (BMD) <u>D) SKY CLIMBER OPERATION & TROUBLE SHOOTING GUIDELINES.</u> 1. Before using Sky Climber, briefing on Safety of Sky Climber to be done. 2. Sky Climber should be operated by trained personnel only. 3. Check operation of individual safety locks every time going on sky climber 4. Check motors are operating normally by going few meters above and coming down before finally going to full height. 5. Stop in case of observance of any abnormality. Come out and inform. 6. Checks motor brakes are working properly on load and are not slipping. 7. Always use PPE's and be attentive while on sky climber. 8. Ensure availability of flexible rope, rope ladder, additional bulb, water, emergency lever for manual operation of motor gear box and torch on sky Climber.		

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9. Don't overload the sky climber. 10. Don't go up until it is operated by experienced operators. 11. Do not try to move up / down more than twice if sky climber is stuck up while in operation. 12. Do not operate safety lock lever while giving command to motor for going up/down. 13. Never keep the safety lock lever in hand. Just release and leave it. 14. Ensure minimum T&P as suggested by OEM /OES. Note: <i>In case of any problem at height, ask for help of competent persons. Do not try to rectify it yourself.</i> Responsibility - HOD (BMD) <u>E) MAINTENANCE AND OVERHAUL OF SKY CLIMBER</u> 1. There shall be periodic inspection of all components of Sky Climber. The frequency of inspection shall be condition based or as suggested in OEM/OES manual, whichever is earlier. 2. Annual rate contract should be made with OEM/OES for, inspection, servicing and load testing of Sky climbers. 3. Training should be imparted by OEM/OES personnel for safe operation of Sky climber to the persons involved in operation and maintenance of sky climber. Training to be provided to sufficient persons to operate minimum 4 sky climbers in parallel. 4. The Sky Climber should be stored carefully to avoid any damage / rusting of its component during storage. Responsibility - HOD (BMD) <u>8.0 RESPONSIBILITIES</u> The responsibility of carrying out different tasks in relation to Safe Operation of Boiler Sky climbers as detailed in the LMI are with HOD (BMD), HOD (SAFETY) & HOD (FQA).		

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<u>9.0 RECORD KEEPING</u> The records to be maintained under this LMI include: a) Compliance record of the activities detailed under this LMI b) A record of the trained persons should be kept at site as per Annexure A c) Protocol as per Annexure B signed by BMD, FQA and Safety department before operation of Sky Climber The above records are to be maintained by HOD (BMD). <u>10.0 REVIEW</u> This document shall be reviewed by Head of The Project on two yearly basis.		

[illegible]

NTPC	13.0 COMPLIANCE RECORD	SIMHADRI
TITLE: GUIDELINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER REV: 02; DATE: DEC'2022		
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UNIT NO.	ACTIVITY	COMPLIANCE

[illegible]

EIC

OEM/OES or Its Representative

Custodian of this annexure: Head of Boiler

NTPC	ANNEXURE-B			SIMHADRI
TITLE: GUIDELINES FOR CARRYING OUT SAFE OPERATION OF SKY CLIMBERS INSIDE BOILER REV: 02; DATE: DEC'2022				
REF NO. LMI/OGN/OPS/MECH/053/		PAGE : 12 OF 13		
Protocol for Sky Climber safety				
Unit No: _____		Date: _____		
Reason For Outage: _____				
The following protocol should be signed by BMD, FQA and Safety department before operation of Sky Climber :				
Sl. No	Checks Before operation of Sky Climber	Type of Check	Observation	Remarks
1	Serial No. of Hoists and Safety Locks being used	Visual	Mention Sl.No	
2	Validity of servicing of Sky Climber by OEM /OES	Visual/ Certificate Verification	Mention Date	
3	One hard copy of OEM/OES manual of sky climber kept in Sky Climber.	Visual	Yes/No	
4	All safety instructions and list of trained persons (Annexure A) are displayed in Sky Climber.	Visual	Yes/No	
5	All persons going on Sky Climber are briefed about Safety of Sky Climber.	Visual	Yes/No	
6	All persons operating sky climber have been trained by OEM/ OES or its representatives	Visual	Yes/No	
7	All persons going on Sky Climber are wearing personnel protective equipment such as helmet, safety belt, nose mask, safety goggles etc. and also have PPE as prescribed by OEM	Visual	Yes/No	
8	Check wire rope physically.	Physical condition	Ok / Not Ok	
9	Check Physical condition of motor & gear	Physical condition	Ok / Not Ok	
10	Check motor brakes are working properly on load and are not slipping	By Actual Checking	Ok / Not Ok	
11	Check all switches and controls of Sky Climber	By Actual Checking	Ok / Not Ok	
12	Check working of anti-tilting device of Sky Climber	Physical condition	Ok / Not Ok	
13	Check emergency lowering lever of Hoists	By Actual Checking	Ok / Not Ok	
14	Check healthiness of safety lock	By Actual Checking	Ok / Not Ok	

(AGENCY)

(BMD)

(FQA)

(SAFETY)

Custodian of this annexure: Head of Boiler

NTPC		ANNEXURE-B		SIMHADRI	
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Protocol for Sky Climber safety					
Sl. No	Checks Before operation of Sky Climber	Type of Check	Observation	Remarks	
15	Check physical condition of platform and other members	Physical condition	Ok / Not Ok		
16	Check edges of guide pipe and clamping of wire rope	Physical condition	Ok / Not Ok		
17	Check dead weight is attached to safety rope and it is hanging	Visual	Yes/No		
18	Check rig wire rope is at least 3-meter excess out of the hoist exit guide	Visual	Yes/No		
19	Check if the power supply is corresponding with the motor power (motor name plate)	Visual	Yes/No		
20	Check that the power supply MUST have an earth-leakage protection device.	By Actual Checking	Ok / Not Ok		
21	Check the roof anchor point arrangement(03 no's U clamp used in wire rope ends)	By Actual Checking	Ok / Not Ok		
22	Check that Sky Climber platform and wire rope are insulated from structure.	Visual	Yes/No		
23	Check that all rubber wall rollers are used in the Sky Climber	Visual	Yes/No		
24	Check that the rubber wall rollers used in the Sky Climbers are in good condition	Physical condition	Ok / Not Ok		
25	Check power supply voltage while hoisting without load.	By Actual Checking	Ok / Not Ok		
25	Check power cable and short cable.	Physical condition	Ok / Not Ok		
26	An additional bulb(24 V), water, ropes of adequate length, additional safety belts, emergency lever for manual operation of motor gear box and torch should be available on the platform.	Visual	Yes/No		
27	Check for adequate light inside the boiler	Visual	Yes/No		
28	Check platform is not overloaded. (Not more than 5 persons)	Visual	Yes/No		
29	Check that there are no obstructions to free vertical travel of the Sky Climber	Visual	Yes/No		
30	Check that provision for firm anchoring of Sky Climber is provided at the level of its use in the boiler	Visual	Yes/No		
31	Check oil level in the gearbox.	By Actual Checking	Ok / Not Ok		
32	Check that the power supply MUST have an earth leakage protection device.(RCCB with 30milli amp capacity)	Yes/No			

(AGENCY) (BMD) (FQA) (SAFETY)

Custodian of this annexure: Head of Boiler

